

Цель работы: сформировать навык настройки VLAN на основе портов.

Выполнение заданий: Вариант 11

Cisco Packet Tracer - C:\Users\minsk\Desktop\Work\4-й курс\Компьютерные сети\ЛР\ЛР 20.pkt

File Edit Options View Tools Extensions Window Help

Logical Physical 337, y: 304

Root 02:10:00

Time: 00:04:49

Realtime Simulation

Scenario 0

New Delete

Toggle PDU List Window

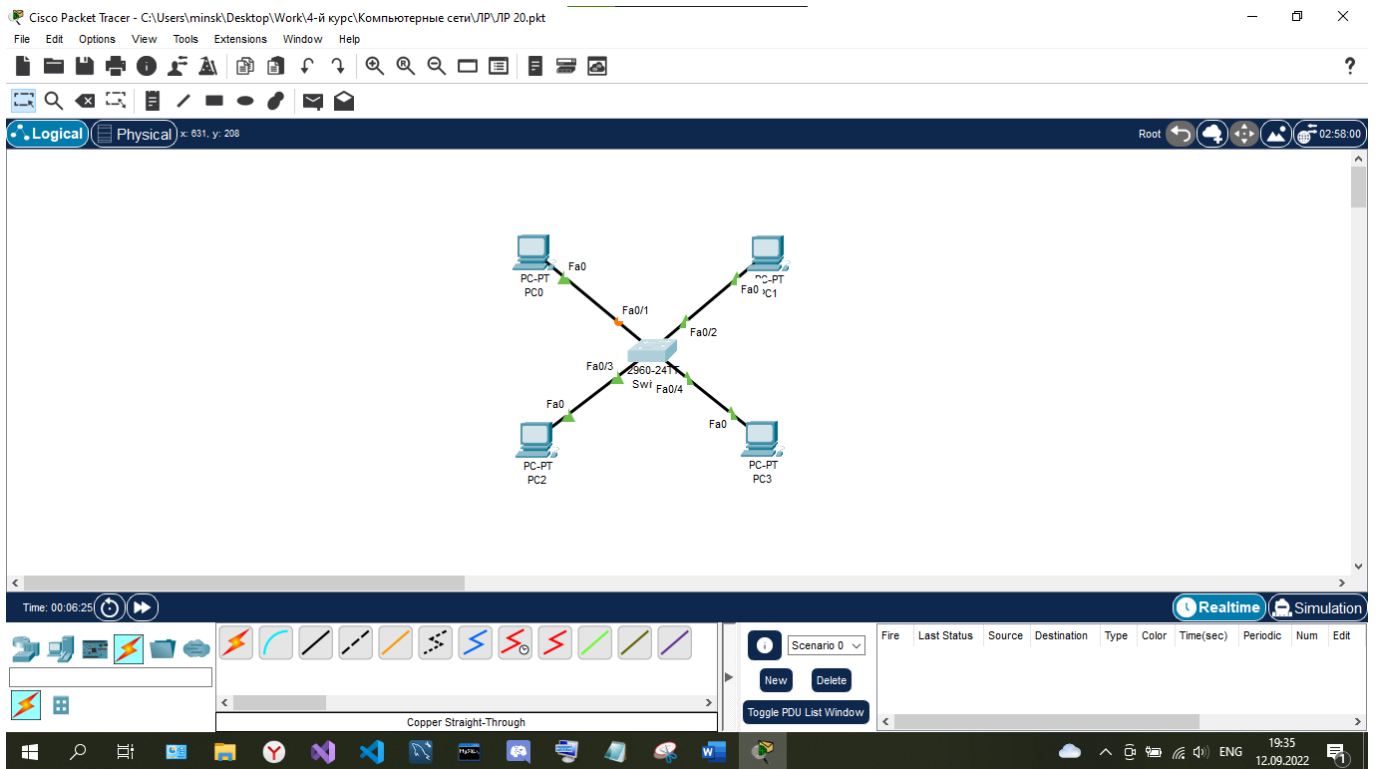
Copper Straight-Through

```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 2
Switch(config-vlan)#name buh
Switch(config-vlan)#exit
Switch(config)#int fa0/1
Switch(config-if)#swi
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 2
Switch(config-if)#exit
Switch(config)#
```

☐ Top

Copper Straight-Through

Toggle PDU Li



```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 2
Switch(config-vlan)#name buh
Switch(config-vlan)#exit
Switch(config)#int fa0/1
Switch(config-if)#swi
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 2
Switch(config-if)#exit
Switch(config)#int fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 2
Switch(config-if)#exit
Switch(config)#
```

☐ Top

Copper Straight-Through

Toggle PDU

Cisco Packet Tracer - C:\Users\minsk\Desktop\Work\4-й курс\Компьютерные сети\ЛР\ЛР 20.pkt

File Edit Options View Tools Extensions Window Help

Logical Physical x: 904, y: 210

Root 04:25:30

Time: 00:09:19

Realtime Simulation

Scenario 0

New Delete

Toggle PDU List Window

Copper Straight-Through

19:38 12.09.2022

Switch0

Physical Config CLI Attributes

IOS Command Line Interface

Switch#show vlan

VLAN	Name	Status	Ports
1	default	active	Fa0/3, Fa0/4, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/1, Gig0/2 Fa0/1, Fa0/2
2	buh	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1	enet	100001	1500	-	-	-	-	-	0	0
2	enet	100002	1500	-	-	-	-	-	0	0
1002	fddi	101002	1500	-	-	-	-	-	0	0
1003	tr	101003	1500	-	-	-	-	-	0	0
1004	fdnet	101004	1500	-	-	-	ieee	-	0	0
1005	trnet	101005	1500	-	-	-	ibm	-	0	0

Remote SPAN VLANs

Primary	Secondary	Type	Ports
Switch#			

Copy Paste

Top

Copper Straight-Through

Toggle PDU List Window

19:39 12.09.2022

```
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 3
Switch(config-vlan)#name users
Switch(config-vlan)#exit
Switch(config)#int fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 3
Switch(config-if)#exit
Switch(config)#
```

Copy

☐ Top

Copper Straight-Through

Toggle PDU List Window

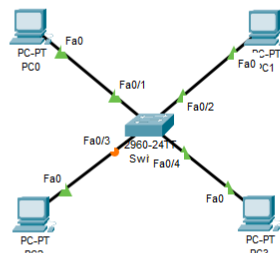
Cisco Packet Tracer - C:\Users\minsk\Desktop\Work\4-й курс\Компьютерные сети\УП\УП 20.pkt

File Edit Options View Tools Extensions Window Help



Logical Physical x: 665, y: 341

Root 05:37:00



Time: 00:11:40



Scenario 0

New Delete

Toggle PDU List Window

Набросок на фрагменте экрана

Фрагмент сохранен в буфере обмена

Нажмите здесь, чтобы оставить пометки на изображении и поделиться им

Cisco Packet Tracer - C:\Users\minsk\Desktop\Work\4-й курс

File Edit Options View Tools Extensions Window

Logical Physical x: 655, y: 199

Time: 00:13:57

Switch0

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 3
Switch(config-vlan)#name users
Switch(config-vlan)#exit
Switch(config)#int fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 3
Switch(config-if)#exit
Switch(config)#int fa0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 3
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24
2 buh	active	Gig0/1, Gig0/2 Fa0/1, Fa0/2
3 users	active	Fa0/3, Fa0/4
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

Copy Paste

Top

Copper Straight-Through

Toggle PDU List Window

Realtime Simulation

Destination Type Color Time(sec) Periodic Num Edit

19:43 12.09.2022

Cisco Packet Tracer - C:\Users\minsk\Desktop\Work\4-й курс

File Edit Options View Tools Extensions Window

Logical Physical x: 785, y: 97

Time: 00:17:52

PC1

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:

Reply from 192.168.2.1: bytes=32 time<1ms TTL=128
Reply from 192.168.2.1: bytes=32 time<1ms TTL=128
Reply from 192.168.2.1: bytes=32 time<1ms TTL=128
Reply from 192.168.2.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.3.1

Pinging 192.168.3.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.3.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Top

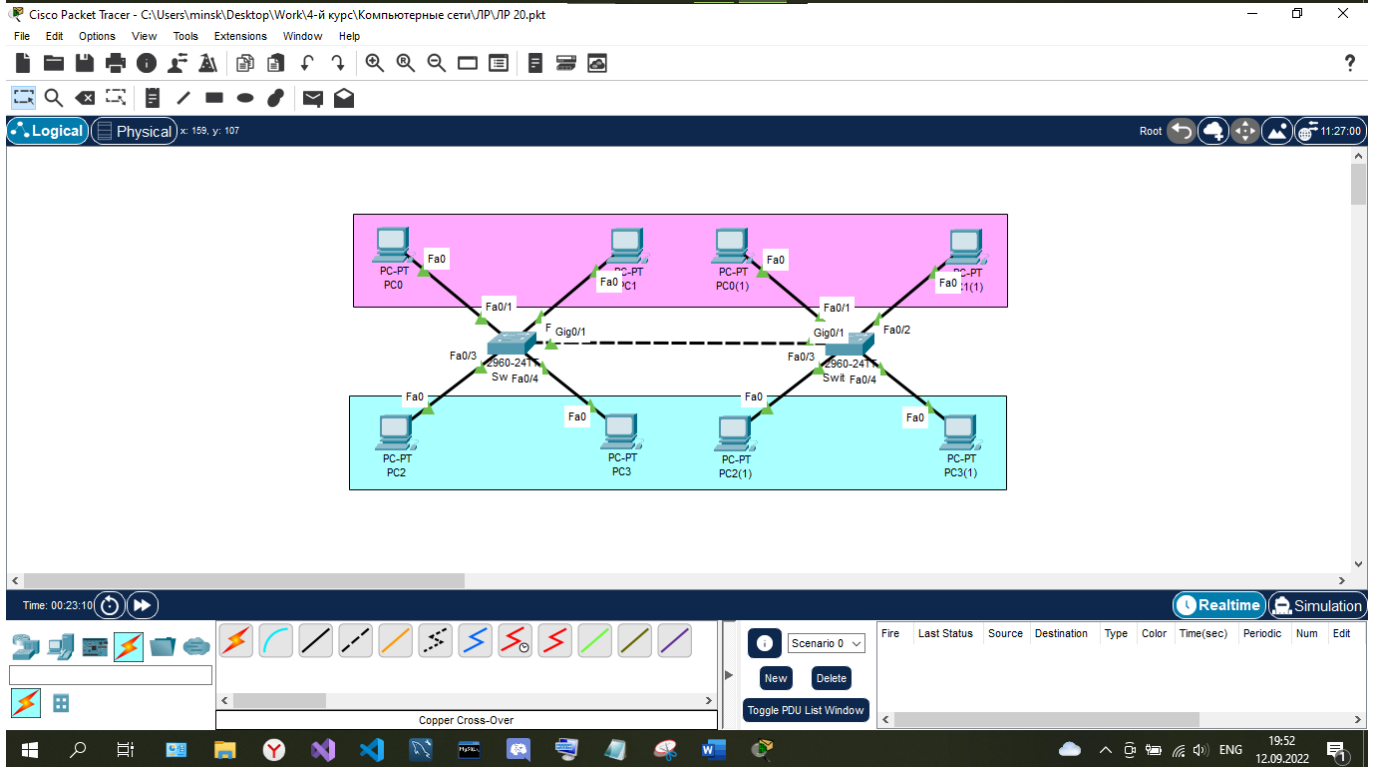
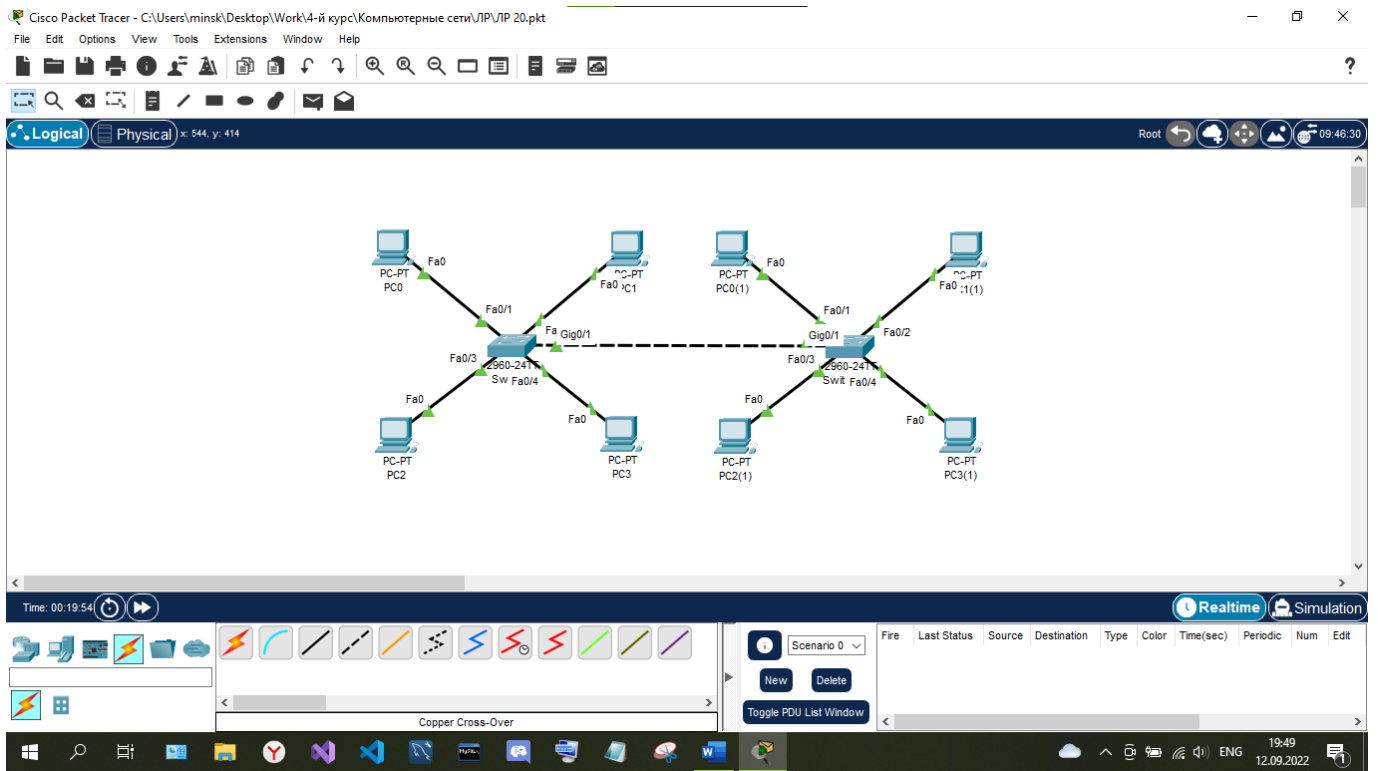
Copper Straight-Through

Toggle PDU List Window

Realtime Simulation

Destination Type Color Time(sec) Periodic Num Edit

19:47 12.09.2022



```

!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
  switchport access vlan 2
  switchport mode access
!
interface FastEthernet0/2
  switchport access vlan 2
  switchport mode access
!
interface FastEthernet0/3
  switchport access vlan 3
  switchport mode access
!
interface FastEthernet0/4
  switchport access vlan 3
  switchport mode access
!
interface FastEthernet0/5
!
--More--

```

☐ Top

Copper Cross-Over

The screenshot shows the Cisco Packet Tracer interface with a configuration window for Switch0 open. The window has tabs for Physical, Config, CLI, and Attributes, with the CLI tab selected. The CLI window displays the following commands and output:

```

Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int gi0/1
Switch(config-if)#swi
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

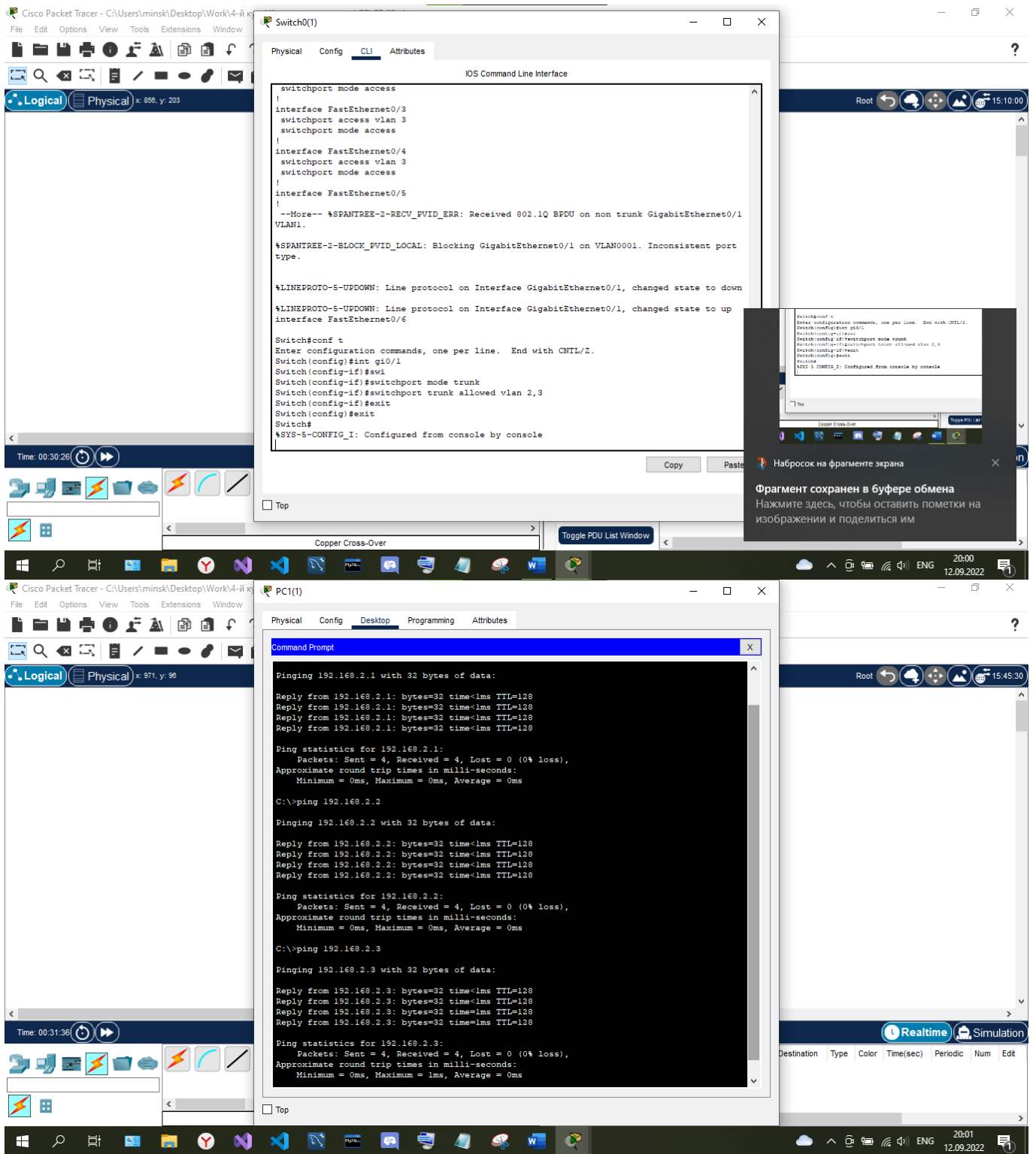
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

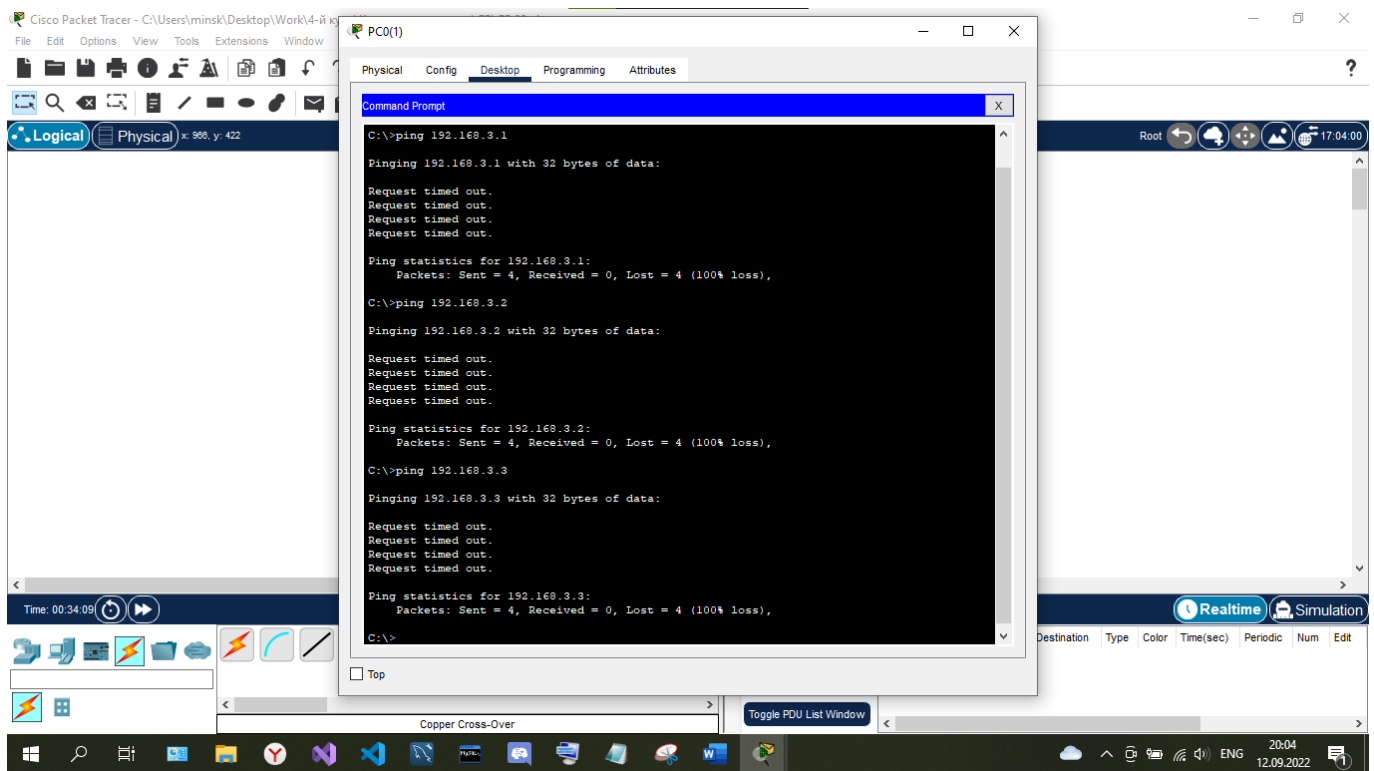
Switch(config-if)#switchport trunk allowed 2,3
^
% Invalid input detected at '^' marker.

Switch(config-if)#switchport trunk allowed vlan 2,3
Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

```

At the bottom of the configuration window, there is a "Top" button. The main Packet Tracer interface shows a network diagram with a "Copper Cross-Over" label and a "Toggle PDU List Window" button. The bottom status bar indicates the time is 19:57 on 12.09.2022, and the language is set to ENG.





Контрольные вопросы

1. Что такое VLAN?

VLAN (Virtual Local Area Network, виртуальная локальная сеть) — это функция в роутерах и коммутаторах, позволяющая на одном физическом сетевом интерфейсе (Ethernet, Wi-Fi интерфейсе) создать несколько виртуальных локальных сетей.

2. В чем преимущества создания виртуальных сетей?

- Гибкое разделение устройств на группы
- Уменьшение широковещательного трафика в сети
- Увеличение безопасности и управляемости сети
- Уменьшение количества оборудования и сетевого кабеля

3. Назовите примеры использования VLAN

- Объединение в единую сеть компьютеров, подключенных к разным коммутаторам.

- Разделение в разные подсети компьютеров, подключенных к одному коммутатору.
- Разделение гостевой Wi-Fi сети и Wi-Fi сети предприятия.

Вывод: я научился настраивать VLAN на основе портов.